17. Windows OS Commands Execution (ping, ifconfig, netstat)

Aim

To execute and understand Windows OS networking commands such as `tracert`, `ping`, `ipconfig` (Windows alternative to `ifconfig`), and `netstat` to check network connectivity, IP configuration, and active network connections.

Procedure

1. Start the Windows Command Prompt by pressing `Win + R`, typing `cmd`, and pressing Enter.
2. Execute the `ping` command to check connectivity to a host.
3. Execute the `tracert` command to trace the route to a destination.
4. Execute the `ipconfig` command to display IP configuration.
5. Execute the `netstat` command to display active network connections.
6. Observe the output of each command.

Code / Commands

```
ping google.com
ipconfig
netstat -an
```

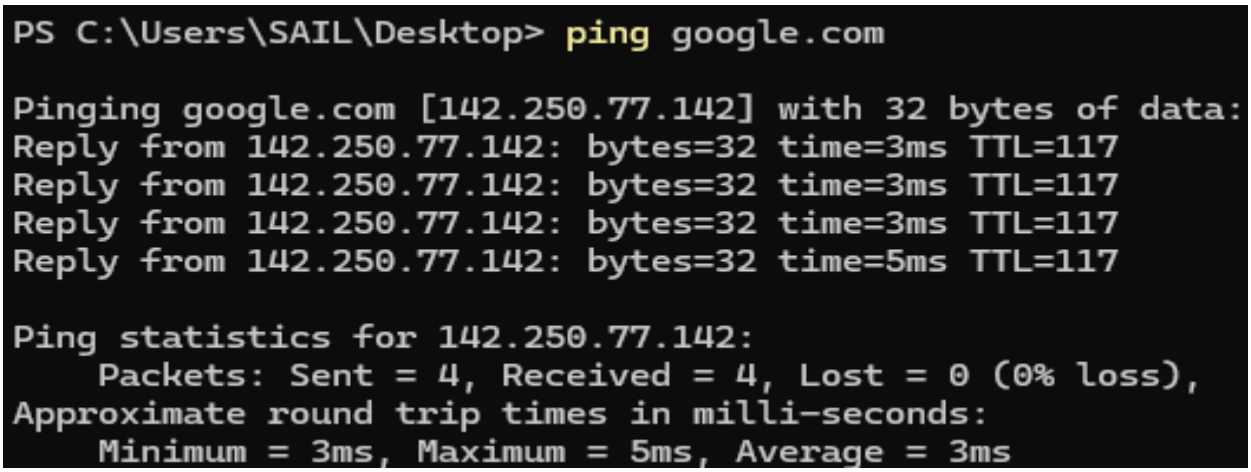
Sample Input and Output

Command 1: ping

Sample Input:

```
ping 127.0.0.1
```

Sample Output:



Command 2: ipconfig

Sample Input:

`ipconfig`

Sample Output:

```
S C:\Users\SAIL\Desktop> ipconfig

indows IP Configuration


thernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::463a:fbd9:ed4d:f303%17
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

thernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::1681:cc3a:bebf:3988%4
   IPv4 Address. . . . . . . . . . . : 172.23.161.112
   Subnet Mask . . . . . . . . . . . : 255.255.240.0
   Default Gateway . . . . . . . . . : 172.23.16.1

thernet adapter VMware Network Adapter VMnet1:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::8a6e:42fd:f052:1699%13
   IPv4 Address. . . . . . . . . . . : 192.168.99.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

thernet adapter VMware Network Adapter VMnet8:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::e31f:9744:7165:e5e5%15
   IPv4 Address. . . . . . . . . . . : 192.168.47.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
```

Command 3: netstat

Sample Input:

`netstat -an`

Sample Output:

```
PS C:\Users\SAIL\Desktop> netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    172.23.161.112:7680    172.19.56.223:52517    ESTABLISHED
  TCP    172.23.161.112:7680    172.23.23.187:57639    ESTABLISHED
  TCP    172.23.161.112:49685   4.213.25.240:https     ESTABLISHED
  TCP    172.23.161.112:52810   lcbomp-in-f188:5228    ESTABLISHED
  TCP    172.23.161.112:56553   SAIL601-PC16:ms-do     ESTABLISHED
  TCP    172.23.161.112:56554   SAIL604-PC07:ms-do     TIME_WAIT
  TCP    172.23.161.112:56555   SAIL604-PC10:ms-do     TIME_WAIT
  TCP    172.23.161.112:57521   40.79.167.9:https      ESTABLISHED
  TCP    172.23.161.112:59124   40.79.167.9:https      ESTABLISHED
  TCP    172.23.161.112:64012   0:https                ESTABLISHED
```

Result

Thus, the Windows networking commands `ping`, `ipconfig`, and `netstat` were studied and executed to check network connectivity, trace routes, view IP configuration, and identify active network connections.